Hotel Booking Cancellation Project

CSE437: Data Science | Section 05 | Summer 2026 | Group 15

A leakage-aware, chronological comparison of Logistic Regression and Random Forest

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Summary

Hotel cancellations complicate revenue planning and room allocation. This project uses the Hotel Booking Demand dataset, containing 119,390 city and resort hotel reservations, to predict the binary target is_canceled. After excluding 180 confirmed zero-guest records, a chronological design reserves 95,415 bookings for development and 23,795 later bookings for testing. Missing-value handling, encoding, feature selection and numeric transformations are fitted only within training partitions. Logistic Regression and Random Forest are compared with a majority baseline; numeric PCA is evaluated but not retained because selection alone performs better in development. Grid search selects balanced Logistic Regression with C=1 and a fixed 0.5 threshold. It achieves test F1 of 0.7506, accuracy of 76.09% and recall of 88.03%. The most important finding is that deposit type has a pronounced association with cancellation, although retrospective recording and repeated booking profiles prevent a causal interpretation. Random Forest and baseline test scores are disclosed as a later reporting supplement, not a new selection exercise. False alerts, weaker short-lead performance and uncalibrated probabilities limit operational use.

Submission status: Recorded evidence supports the analysis, with numerical reproduction limitations disclosed. The original data, model and numerical results are preserved. Repository reorganization changes paths and metadata only; current notebook execution and final author review remain to be verified.

1. Problem and Dataset

1.1 Problem statement

Hotel booking cancellations can cause loss of money and make room planning difficult for hotels. The goal of this project is to study the factors that affect booking cancellations and build a machine-learning model that can predict whether a hotel booking will be canceled.

1.2 Dataset

The supplied Hotel Booking Demand CSV has 119,390 rows, 32 columns and 16,855,599 bytes, covering arrivals from 1 July 2015 to 31 August 2017. It contains 79,330 City Hotel and 40,060 Resort Hotel records. The Kaggle distribution by Jesse Mostipak acknowledges earlier preparation by Thomas Mock and Antoine Bichat [1]. This project uses that supplied combined CSV; it does not scrape or merge another dataset.

Kaggle's public metadata lists the dataset licence as CC BY 4.0 [1]. The current metadata version is 1; the original acquisition date and acquired version remain unknown, as confirmed by the user. The licence information, unchanged source checksum and acquisition instructions are in data/README.md. The original CSV is publicly committed; its Git blob matches the checksum-verified source. Its size is below the faculty's 50 MB threshold.

1.3 Target variable

The binary target is is_canceled: 1 means canceled and 0 means not canceled. The raw counts are 44,224 canceled and 75,166 not canceled (37.04% versus 62.96%). Excluding 180 confirmed zero-total-guest rows leaves 119,210 bookings. Development contains 34,473 cancellations among 95,415 bookings (36.13%); the later test contains 9,726 among 23,795 (40.87%). Development-only statistics informed modeling.

1.4 Three questions

1. Which booking and customer-related factors have the biggest effect on hotel cancellations?

2. How accurately can machine-learning models predict whether a hotel booking will be canceled?

3. Which machine-learning model gives the best result after data preprocessing, feature selection, dimensionality reduction, and hyperparameter tuning?

The wording is unchanged. Lead time, deposit type and previous cancellations were the predeclared focus. "Effect" is interpreted as association, not causation.

2. Data Handling and Preprocessing

2.1 Data quality audit

Raw-data issue	Recorded evidence
Missing values	children 4; country 488; agent 16,340; company 112,593
Other columns	No missing values detected
Exact duplicate copies beyond the first	31,994; retained because no booking ID resolves genuine repeats
Guest totals	180 confirmed zeros; 4 unknown totals; no negative guest counts
Explicit Undefined categories	meal 1,169; market_segment 2; distribution_channel 5
ADR	Minimum -6.38; maximum 5,400; one negative value

Both reservation_status and reservation_status_date encode outcomes and are excluded from predictors. Undefined category values remain distinct from missing values. The original full-source quality audit preceded splitting; this inspection is disclosed rather than described as an entirely untouched-data workflow.

2.2 Missing values

Mechanisms are unverified. Agent/company NULLs are treated as potentially structural, for example when no agent or company is associated with a booking; this is a working interpretation, not a confirmed missingness mechanism. Country/children/invalid ADR have unknown mechanisms; missing completely at random is not assumed.

Field	Treatment and rationale
children / numeric missingness	Training median + children indicator; zero fallback for entirely missing training columns.
country	Explicit Unknown category; no country is fabricated.
agent	NoAgent for missing values; nominal string codes.
company	Drop sparse ID (94.31% missing); derive presence flag.
adr	Negative to missing; training median + indicator.
Derived totals/share	Propagate unknowns; training medians; indicators for total nights, total guests and cancellation share.

2.3 Outliers

A raw ADR 1.5-IQR diagnostic (Q1=69.29, Q3=126; bounds -15.775/211.065) flagged 3,793 values, but missed the single negative ADR. These bounds were diagnostic, not deletion thresholds. Domain rules turn the negative value into missing; 1,959 raw zero-ADR values and high positive values, including 5,400, are retained. Only 180 confirmed zero-guest rows are excluded.

2.4 Transformation and scaling

After imputation, log1p transforms lead time, prior canceled/noncanceled bookings, ADR and three derived totals. LR uses StandardScaler; RF is unscaled. One-hot encoding maps unseen categories to an all-zero block. The split precedes learned preprocessing: imputers, scalers, vocabularies, selectors and PCA fit training partitions only.

Training ADR medians are 76.50/80.75/90.00; fold 3's negative validation ADR uses 90.00. Encoded widths are 328/422/491 with zero nonfinite outputs and matching train/validation width. Assigned room, booking changes and waiting-list days are excluded as potentially updated fields.

2.5 Before and after

Stage	Rows	Predictor fields / encoded width
Raw CSV	119,390	32 total columns
Separate target and two status fields	119,390	29 candidate predictors
Remove confirmed zero guests	119,210	29 candidates
Initial source-field policy	119,210	25 source fields
Derived-feature schema	119,210 eligible	32 fields before encoding
Selected final LR representation	95,415 fitted; 23,795 tested	406 encoded features

3. Statistical Analysis

3.1 Descriptive statistics

Development field	Mean	Median	Q1-Q3	Maximum
Lead time (days)	96.59	60	15-145	737
ADR, valid observed values	93.71	88	65-115	5,400
Previous cancellations	0.106	0	0-0	26
Weekday nights	2.45	2	1-3	50

Lead time and ADR are right-skewed; prior cancellations concentrate at zero. Development has 62,827 City and 32,588 Resort bookings. Deposit frequencies are 82,979 No Deposit, 12,296 Non Refund and 140 Refundable. Full frequency/spread tables are in development EDA.

3.2 Relationships

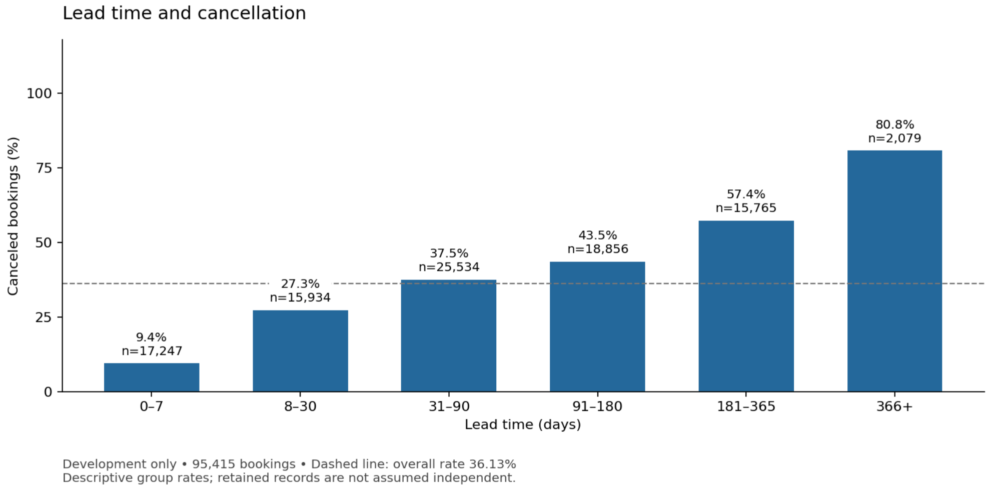

Figure 1. Cancellation by fixed lead-time bands; development only.

Longer lead-time bands have progressively higher cancellation rates: 9.43% for 0-7 days (n=17,247) versus 80.81% for 366+ days (n=2,079). The direction is consistent within both hotels.

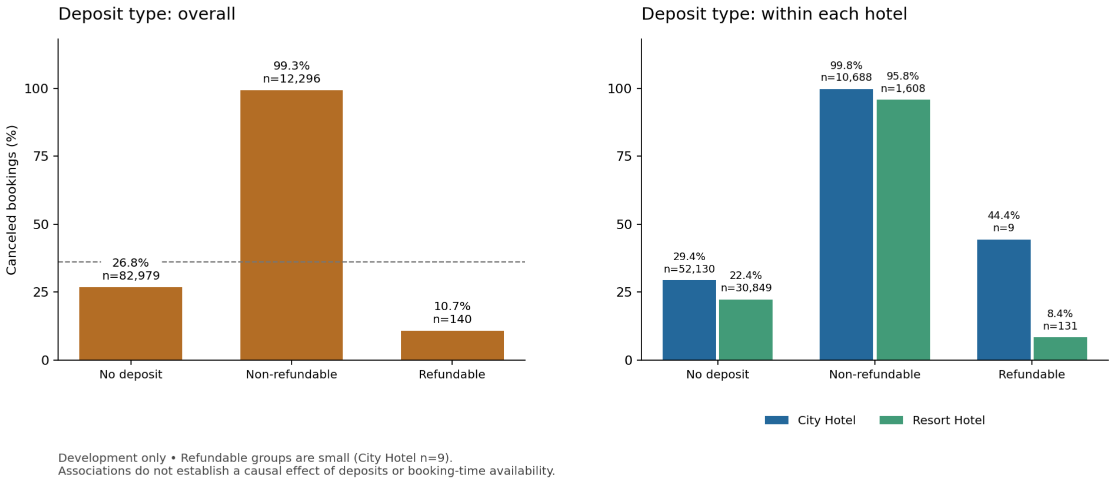

Figure 2. Deposit category differences and hotel stratification; development only.

Non Refund has 99.25% cancellation, compared with 26.82% for No Deposit and 10.71% for Refundable. The latter has only 140 observations. Booking mix, policies and retrospective recording may contribute; the figures do not show that deposits cause cancellation.

3.3 What the data says so far

- Preserve lead time and deposit type as planned predictors; assess their limitations rather than infer causality.
- Prior-cancellation rates are non-monotonic: 32.07%, 95.65%, 31.29% and 74.77% for counts 0, 1, 2-3 and 4+. The last two groups have only 163 and 222 records.
- City Hotel cancellation is 41.41%, versus 25.94% for Resort Hotel; pooled patterns depend on hotel mix.
- Equal-profile weighting changes overall cancellation from 36.13% to 25.26%, and the 4+ prior-cancellation rate from 74.77% to 26.32%. Repetition sensitivity is substantial.
- Retain grouped, chronological validation. Do not use naive independent-row confidence intervals or describe EDA rates as model importance.

Equal-profile weighting averages conflicting labels within each of 67,961 candidate-profile groups. It describes the average profile rather than the average booking; it is a sensitivity analysis, not a replacement training population.

4. Feature Engineering

4.1 Derived features

Constructed field(s)	Definition and reason
total_nights	Weekend plus weekday nights; booking duration.
total_guests	Adults plus children plus babies; party size.
previous_bookings_total	Previous canceled plus noncanceled bookings; history volume.
has_booking_history	Indicator of positive history total; distinguishes no history.
previous_cancellation_share	Canceled/history total; zero for no history, interpreted with the history flag.
company_code_recorded	Presence of a company code; recording proxy, not proof of corporate payment.
arrival_month_sin / arrival_month_cos	sin/cos of 2*pi*(month-1)/12; cyclic seasonality, replacing month names.

These eight deterministic fields extend 24 retained source fields to 32 inputs. Unknown components propagate before imputation. The audit retains 604 zero-night development bookings and four unknown guest totals. Separately imputed totals need not equal sums of separately imputed components. Ratios, flags and calendar coordinates are not logged.

4.2 Dimensionality reduction

Centered full-SVD PCA retains at least 95% training variance in the scaled numeric block; category and missing-indicator columns bypass PCA. It reduces 23 numeric fields to 16/15/16 components across folds, retaining 95.88%/95.00%/95.69% variance. Selection-plus-PCA uses 15 components per fold. PCA is demonstrated but not retained: preserving variance did not improve the agreed development F1 over selection alone.

4.3 Feature selection

Training-constant encoded columns (variance at most 1e-12) are removed. Remaining columns are ranked by training-label ANOVA F; the top 75%, rounded upward, are retained, with ties resolved by encoded order. Scores are ranking heuristics, not causal importance or inferential p-values.

Fixed representation with reference LR	Mean development F1	Fold output widths
All engineered features	0.693094	332 / 421 / 490
Selection only	0.713609	247 / 314 / 366
Numeric PCA	0.694633	325 / 413 / 483
Selection then PCA	0.701023	242 / 308 / 360

4.4 Final feature set

The 32 pre-encoding inputs are: lead_time; previous_cancellations; previous_bookings_not_canceled; adr; total_nights; total_guests; previous_bookings_total; arrival_date_year; arrival_date_week_number; arrival_date_day_of_month; stays_in_weekend_nights; stays_in_week_nights; adults; children; babies; is_repeated_guest; required_car_parking_spaces; total_of_special_requests; has_booking_history; previous_cancellation_share; company_code_recorded; arrival_month_sin; arrival_month_cos; hotel; meal; country; market_segment; distribution_channel; reserved_room_type; deposit_type; agent; customer_type.

The final training-fitted selection retains 406 encoded features, listed completely in feature_coefficients.csv. The rule, not a global pre-CV mask, is refitted within each training partition. Dropped fields are the two outcome-status columns, raw company ID, assigned room, booking changes, waiting-list days and replaced month names. Selection is justified by the development comparison, not test performance; no isolated benefit is claimed for every engineered field.

5. Modeling and Validation

5.1 Validation strategy

The split uses whole arrival dates near an 80/20 row target: 95,415 development bookings (80.04%) end on 22 April 2017, and 23,795 test bookings begin on 23 April. Three expanding forward folds remain entirely within development. Boundaries are deterministic and do not use target labels; no random shuffle or label stratification is applied. Model seeds are 42.

Fold	Training rows / end	Validation rows / period
1	23,797 / 26 Jan 2016	23,893 / 27 Jan-21 Jun 2016
2	47,690 / 21 Jun 2016	23,776 / 22 Jun-6 Nov 2016
3	71,466 / 6 Nov 2016	23,949 / 7 Nov 2016-22 Apr 2017

Candidate-profile groups cannot cross development/test or training/validation within a fold. This guards recorded duplicate profiles, not unobserved shared guests. There is no temporal embargo, and arrival cohorts do not establish real-time label availability. The design is retrospective, not a simulated live booking system.

5.2 Baseline

A training-majority DummyClassifier predicts not canceled. Its mean forward-validation F1 and recall are zero, despite 63.82% accuracy. This shows why majority accuracy is not an adequate success criterion.

5.3 Model families

Logistic Regression supplies a regularized linear log-odds baseline for encoded features; nonlinear relations require suitable transformations, and coefficients are not causal. Random Forest models nonlinear splits and interactions without numeric scaling, but can overfit repeated patterns and transfer poorly across time. Both full and selected representations are compared before tuning.

Untuned candidate	Mean F1	Precision	Recall
Logistic Regression, full	0.693094	0.681025	0.753386
Logistic Regression, selected	0.713609	0.783364	0.659498
Random Forest, full	0.626504	0.906662	0.481119
Random Forest, selected	0.657994	0.893717	0.521714

5.4 Metrics

The split plan fixes cancellation-class F1 as primary, aggregated by the unweighted mean across three folds. F1 balances precision and recall under moderate class imbalance. Accuracy, precision, recall and ROC-AUC provide context; Brier score is a later probability diagnostic. Undefined precision is reported as zero. Threshold remains 0.5. No measured business costs justify optimizing financial savings.

6. Hyperparameter Tuning

6.1 Search space

Family	Searched hyperparameter	Values
Logistic Regression	C	0.01, 0.1, 1, 10
Logistic Regression	class_weight	None, balanced
Random Forest	max_depth	8, 16, None
Random Forest	min_samples_leaf	1, 10
Random Forest	class_weight	None, balanced

6.2 Method

GridSearchCV evaluates 8 LR and 12 RF candidates over 3 frozen folds: 60 fits. Each complete pipeline refits preprocessing/selection on training only. LR uses lbfgs and max_iter=2000; RF uses 100 trees and sqrt feature sampling. Both use seed 42, selected representation and threshold 0.5. Search refit=False defers full-development fitting until selection is frozen.

6.3 Results

Family and weights	Search trend: mean development F1
LR, unweighted; C=0.01 / 0.1 / 1 / 10	0.7012 / 0.7102 / 0.7136 / 0.7147
LR, balanced; C=0.01 / 0.1 / 1 / 10	0.7267 / 0.7317 / 0.7321 / 0.7304
RF, balanced, leaf=10; depth=8 / 16 / None	0.5796 / 0.6543 / 0.6693
RF, unweighted, leaf=10; depth=8 / 16 / None	0.4972 / 0.5590 / 0.5892

Best family configuration	Untuned F1	Tuned F1	Change
LR: C=1, balanced weights	0.713609	0.732102	+0.018492
RF: depth=None, leaf=10, balanced weights	0.657994	0.669294	+0.011300

Balanced LR improves recall from 0.6595 to 0.7568 while precision falls from 0.7834 to 0.7153. Stronger/deeper models are not uniformly better: balanced LR declines at C=10; unrestricted RF with leaf=1 shows a larger training-validation gap than the selected leaf=10 forest. Full 20-candidate results, fold scores and settings are in tuning evidence.

The selected LR setting is best among the evaluated choices, not globally optimal. Reusing the same development folds for feature, family and parameter decisions creates selection optimism; no significance claim is made. All 60 fits completed without convergence failures. The selected family, parameters and threshold were frozen before evaluation testing.

7. Results, Visualization and Error Analysis

7.1 Test set performance

The selected LR pipeline was fitted on all 95,415 development rows and evaluated on 23,795 later bookings. verification subsequently added the development-selected RF and baseline with user approval. This late addition occurred after LR test results were known; it is not a preregistered simultaneous three-model test. No model, feature or threshold was reselected. Saved LR predictions and fitted model remain unchanged.

Model	F1	Accuracy	Precision	Recall	ROC-AUC	Brier
Majority baseline	0.000000	0.591259	0.000000	0.000000	0.500000	0.408741
Logistic Regression, selected	0.750592	0.760874	0.654187	0.880321	0.875977	0.160007
Random Forest, supplement	0.723115	0.789325	0.781239	0.673041	0.878190	0.145476

Model	TN	FP	FN	TP
Majority baseline	14,069	0	9,726	0
Logistic Regression	9,543	4,526	1,164	8,562
Random Forest	12,236	1,833	3,180	6,546

All rows use identical test membership and threshold 0.5. LR has higher F1/recall; RF has higher precision/accuracy and slightly higher ROC-AUC. Lower Brier error does not establish calibration. The baseline predicts no cancellations, so precision is zero by convention. A test F1 above the development mean is a period-specific outcome, not proof of improvement.

7.2 Visualization

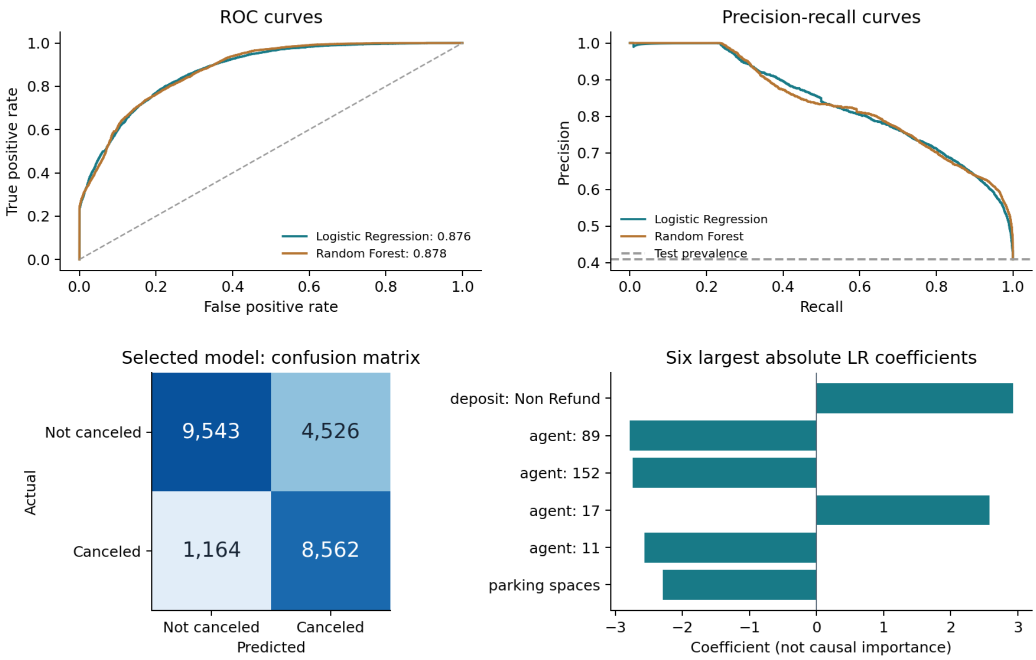

Figure 3. Frozen held-out prediction diagnostics; RF curves are the disclosed late supplement.

The ROC and precision-recall curves expose discrimination and operating-point tradeoffs; they do not select a new threshold. The confusion matrix shows LR's 8,562 detected cancellations, 1,164 missed cancellations and 4,526 false alerts. Non Refund has the largest positive coefficient (+2.929658). Several large coefficients are agent categories. Different encoding/scaling and correlated inputs prevent a universal raw-feature importance ranking.

7.3 Error analysis

LR subgroup	Recorded weakness or contrast
City / Resort Hotel	Error rates 25.59% / 20.24%
Online TA	31.95% error; 3,653 false positives
Lead time 0-7 / 8-30 days	F1 0.3344 / 0.5882
Non Refund	2,290 of 2,291 test bookings canceled
Probability bins	Observed rates below mean predicted probabilities in every fixed bin

Most errors occur in No Deposit bookings. The weighted LR outputs should not be presented as calibrated probabilities. Subgroups are descriptive post-test diagnostics, not grounds for retrospective tuning.

Two concrete errors illustrate what the recorded predictors cannot reliably resolve:

Source row ID	Actual / predicted; probability	Why this case is difficult
14,182	Not canceled / canceled; 0.999603	Direct, No Deposit, lead time 80 days, one prior cancellation. A strong score did not determine this customer's outcome; combinations of historical associations cannot recover unrecorded intent.
94,387	Canceled / not canceled; 0.006030	Lead time 1 day, Complementary segment, Transient-Party, four prior cancellations and three special requests. The rare combination received a low score despite cancellation history; the actual reason for cancellation is not observed.

These are plausible interpretations of model difficulty, not known causal explanations. Twenty distinct confident and near-threshold errors, full group metrics and probability-bin diagnostics are preserved in evaluation evidence.

7.4 Answers to the three questions

Question 1: Which booking and customer-related factors have the biggest effect on hotel cancellations?

Among the three predeclared factors, deposit type shows the sharpest descriptive separation: Non Refund cancellation is 99.25% versus 26.82% for No Deposit in development. Lead-time rates increase from 9.43% to 80.81% across the shortest/longest fixed bands, including within both hotels. Prior-cancellation rates are non-monotonic and sensitive to repeated-profile weighting. The positive Non Refund coefficient is consistent with its association, but this evidence cannot establish causal effects or an exhaustive importance ranking of all predictors.

Question 2: How accurately can machine-learning models predict whether a hotel booking will be canceled?

The selected LR achieves test F1 0.750592, accuracy 76.09%, precision 65.42%, recall 88.03% and ROC-AUC 0.875977. It detects most cancellations, but falsely flags 4,526 noncancellations and misses 1,164 cancellations. Short-lead and Online TA cases are weaker. This is useful retrospective discrimination within the measured source/period, not guaranteed performance for other hotels, calibrated risk or measured financial benefit.

Question 3: Which machine-learning model gives the best result after data preprocessing, feature selection, dimensionality reduction, and hyperparameter tuning?

Selected-feature LR with C=1 and balanced weights is best evaluated under the predeclared development F1 objective: 0.732102 versus 0.669294 for the best RF. Selection beats the full representation with the reference LR; numeric PCA and selection-plus-PCA do not beat selection alone, so PCA is not in the final pipeline. The late test comparison also reports higher LR F1, but it does not determine model choice. RF is better on some secondary test metrics. No universal superiority or global optimum is claimed.

The supporting methods and the preserved representation, tuning and evaluation tables provide the detailed question-to-evidence mapping.

8. Limitations and Next Steps

- Retrospective source timing remains uncertain. Removing direct status leakage and update-prone fields does not prove all retained values were available when a booking decision would be made.
- The full-source quality audit preceded the split. Arrival-based chronology does not guarantee live label availability; no temporal embargo or external-hotel validation was performed.
- Repeated profiles can dominate booking-weighted results. Group separation guards recorded profiles only; sparse categories and unknown customer identity limit statistical independence claims.
- One later-arrival holdout has partial seasonal coverage and higher cancellation prevalence. Reused development folds, finite search and reference-LR-only PCA comparisons limit generalization and claims of improvement.
- The supplementary RF/baseline test comparison was authorized after LR results were known. Its timing is disclosed, and no choice was changed in response.
- Model outputs are not calibrated probabilities; costs of false alerts and misses are not measured. Automated overbooking or production deployment is not justified.
- Numerical reproduction is incomplete: the Windows development rerun selected balanced LR with C=0.1 rather than the original C=1. The numerical cause is unconfirmed. Rerun evidence is isolated; the published selection and cached test results were not replaced or retrained.

Future work should first verify predictor availability at the intended decision time. Deposit ablation, calibration, cost-sensitive thresholds and evaluation across additional hotels/time periods require a new predeclared design with fresh evaluation data. These are proposed experiments, not completed improvements.

9. Contributions

The group confirms that both members completed their assigned responsibilities below. This is a member-reported contribution record, not an independent authorship audit.

Member	Student ID	Contribution record
Faraaz Jamil Chowdhury	24241205	Planning, dataset preparation and provenance, data audit, cleaning/preprocessing, split design, descriptive statistics and EDA.
Ihfaz Rashid Sadat	23301499	Feature engineering, selection/PCA, model comparison and tuning, evaluation/error analysis, reproduction checks and report assembly.

Joint review: Both members are responsible for reviewing, comparing and explaining the whole project. Both accounts appear in the repository history; account identity alone does not establish personal authorship. Final joint sign-off remains open.

References

[1] Mostipak, J. Hotel booking demand. Kaggle dataset. Original dataset authors: Antonio, N., de Almeida, A., and Nunes, L. Metadata credits earlier preparation by Thomas Mock and Antoine Bichat for TidyTuesday. Public dataset metadata was verified on 2 September 2026: CC BY 4.0; current version 1. Original acquired version/date unknown. Raw bytes are unchanged; project preprocessing is documented separately. No endorsement by the original authors or distributor is implied.

AI assistance declaration

ChatGPT assisted with project planning and work division. Antigravity assisted with diagnostic and verification runs.

Declaration status: Author-supplied provisional wording; final review pending.

Reproducibility and submission status: The prior Windows verification passed dependency checks, 70 tests and all five fresh-kernel notebooks with zero cell errors; cached final results were verified, not refitted. Development tuning differed: C=1 balanced mean F1 changed from 0.732102 to 0.731371, below C=0.1 balanced at 0.731697. The archive does not establish the complete checkout's exact commit. Repository cleanup preserves numerical evidence, rebases integrity manifests and retains pre-cleanup digests; it is not a new execution. Colab setup can require a session restart after installation. The reorganized notebooks still require fresh-kernel verification with saved outputs. See methods and verification. Final declaration review and joint author sign-off remain open.